Machine Learning-Based Earthquake Magnitude Prediction with Risk-Based SMS Alert System

Potla Sai Sujith- 2420030280

R. Rahul - 2420090043

NikhileswarReddy - 2420030596

Sumith Reddy - 2420030782

Under the Esteemed Guidance of

Dr. K Swanthana , Assistant Professor, Department of Computer Science and Engineering

KLH CSE

Department of Computer Science and Engineering

KLH University, Hyderabad, India

ABSTRACT

Earthquakes are among the most destructive natural hazards and can cause significant loss of life, infrastructure damage, and economic disruption. Rapid identification of earthquake characteristics and timely dissemination of risk information are therefore important components of disaster preparedness. Traditional earthquake analysis methods often rely on statistical and seismological techniques that may have limitations when dealing with complex nonlinear relationships among geographical and seismic parameters.

This research proposes a Machine Learning-Based Earthquake Magnitude Prediction with Risk-Based SMS Alert System that combines machine learning-based magnitude estimation with an automated location-aware warning mechanism. Historical earthquake records are collected from publicly available seismic datasets and processed through data cleaning, missing-value handling, feature selection, normalization, and transformation. Features such as latitude, longitude, depth, occurrence time, and relevant seismic measurements are used to train and evaluate regression models including Linear Regression, Decision Tree Regressor, Random Forest Regressor, XGBoost, and Artificial Neural Network models. The models are evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and coefficient of determination (R^2).

The predicted magnitude is subsequently converted into predefined earthquake-risk categories. Based on the predicted magnitude and geographical coordinates, the proposed system determines whether registered users fall within a configurable alert radius. When the predicted earthquake risk satisfies the specified threshold, the system generates an SMS notification through the Twilio communication platform. The system operates in simulation mode during development to avoid unintended message delivery. The proposed framework integrates prediction, risk classification, geographical filtering, and communication into a single architecture, providing a practical bridge between machine-learning-based earthquake analysis and disaster-alert mechanisms.

Keywords: Earthquake Prediction, Machine Learning, Magnitude Prediction, Random Forest, XGBoost, Artificial Neural Network, Risk Classification, SMS Alert, Disaster Management, Twilio.

1. INTRODUCTION

1.1 Background

Earthquakes occur because of the sudden release of energy within the Earth's crust, producing seismic waves that propagate through surrounding regions. Depending on their magnitude, depth, location, and proximity to

populated areas, earthquakes can result in casualties, structural damage, landslides, fires, infrastructure failures, and economic losses.

Earthquake monitoring traditionally depends on seismic stations and specialized instruments that continuously measure ground motion. After an earthquake occurs, seismological systems estimate important parameters including magnitude, depth, epicenter, and geographic coordinates. The increasing availability of historical seismic datasets provides an opportunity to investigate machine learning techniques for identifying relationships between earthquake characteristics.

Machine learning algorithms are capable of learning patterns from historical data and can model relationships between multiple input variables and a target variable. In earthquake-related research, machine learning has been investigated for earthquake detection, magnitude estimation, seismic signal classification, aftershock analysis, and risk assessment.

The objective of this research is not to claim deterministic prediction of exactly when and where an earthquake will occur. Instead, the proposed system focuses on machine-learning-based estimation of earthquake magnitude from available earthquake characteristics, followed by risk classification and automated communication.

1.2 Problem Statement

Existing earthquake analysis systems primarily concentrate on detecting seismic activity and estimating earthquake parameters. Machine learning research has also explored earthquake magnitude estimation using different algorithms and datasets. However, prediction models and user-facing emergency communication systems are frequently treated as separate components.

The problem addressed by this research is: How can a machine-learning-based earthquake magnitude estimation model be integrated with geographical risk classification and an automated SMS notification mechanism to provide timely, configurable earthquake-risk alerts?

The proposed system receives earthquake-related parameters, predicts or estimates magnitude using trained machine learning models, categorizes the predicted magnitude into a risk level, determines the geographical region affected using latitude, longitude, and a configurable radius, and generates SMS alerts for registered users when the predefined risk conditions are satisfied.

1.3 Research Gap

- Many studies concentrate primarily on improving earthquake prediction or magnitude estimation accuracy, while practical communication of model outputs receives comparatively less attention.
- Different machine learning algorithms are often evaluated independently; systematic comparison can provide useful evidence for model selection.
- Earthquake-related machine learning systems may focus on seismic datasets without integrating prediction with a complete alert workflow.
- A practical alert system must consider magnitude thresholds and geographical proximity rather than sending a generic notification to every registered user.
- The proposed research combines prediction, risk classification, geographical filtering, and SMS communication.

1.4 Research Objectives

- To develop a machine learning model for earthquake magnitude estimation.
- To preprocess and analyze historical earthquake datasets.

- To compare multiple machine learning algorithms for magnitude prediction.
- To evaluate models using MAE, MSE, RMSE, and R^2.
- To classify earthquakes according to predefined magnitude-based risk levels.
- To incorporate latitude and longitude into the alert decision process.
- To develop an SMS-based notification mechanism.
- To notify registered users located within a configurable geographical radius.
- To provide a practical framework connecting machine learning predictions with disaster-risk communication.
- To maintain a simulation mode during development for safe testing of SMS functionality.

1.5 Research Questions

- RQ1: Which machine learning algorithm provides the most suitable performance for earthquake magnitude estimation?
- RQ2: How do geographical and seismic features influence magnitude estimation?
- RQ3: Can predicted magnitude be effectively converted into predefined earthquake-risk categories?
- RQ4: Can geographical proximity be incorporated into an automated earthquake-alert mechanism?
- RQ5: Can machine-learning predictions be integrated with SMS communication to provide a practical disaster-alert workflow?

1.6 Contributions

- A machine-learning-based earthquake magnitude estimation framework.
- Comparative evaluation of multiple regression algorithms.
- A magnitude-based earthquake risk classification mechanism.
- Integration of geographical coordinates and configurable alert radius.
- Automated SMS alert generation for registered users.
- Simulation mode for safe testing of the notification system.
- A complete workflow connecting earthquake data, prediction, risk assessment, and communication.

1.7 Organization of the Paper

The remainder of this paper is organized as follows. Section 2 discusses related earthquake prediction and machine learning approaches. Section 3 presents the proposed methodology, dataset, preprocessing, feature engineering, models, and alert algorithm. Section 4 describes the experimental setup and evaluation metrics. Section 5 presents the experimental results. Section 6 discusses the findings and practical implications. Section 7 describes limitations. Section 8 concludes the research and presents future work.

2. RELATED WORK

2.1 Existing Approaches

Traditional earthquake analysis depends heavily on seismological instruments and mathematical models. Earthquake magnitude is generally determined from measurements of seismic waves and ground motion. These techniques are highly important for operational earthquake monitoring. However, conventional approaches may not capture every nonlinear relationship that exists among earthquake-related variables. Machine learning provides an alternative approach by learning patterns from historical observations.

2.2 Machine Learning Approaches

Linear Regression establishes a linear relationship between input features and earthquake magnitude: y-hat = beta-zero + beta-one x-one + ... + beta-n x-n. It provides a useful baseline because it is simple and interpretable.

Decision Tree Regression recursively divides the feature space into regions and generates a prediction for each region. It can model nonlinear relationships without requiring a mathematical assumption of linearity.

Random Forest combines multiple decision trees and aggregates their predictions. It can model nonlinear interactions and is relatively robust to variations in training data.

XGBoost is a gradient-boosting method that sequentially builds decision trees, with each new tree attempting to reduce errors produced by previous trees. Its ability to model nonlinear relationships makes it suitable for structured earthquake datasets.

Artificial Neural Networks consist of interconnected computational units arranged in layers. ANNs can model complex nonlinear relationships between earthquake characteristics and magnitude.

2.3 Deep Learning Approaches

Deep learning techniques such as CNNs, LSTMs, and Transformer architectures have increasingly been applied to seismic waveform and spatio-temporal data. LSTM models are relevant when temporal dependencies are important, CNN models can learn spatial or signal-level patterns, and Transformer-based approaches can model long-range dependencies. The proposed system primarily operates on structured earthquake parameters rather than raw seismic waveforms.

2.4 Explainability / Existing Techniques

Machine learning models used in disaster-related applications should ideally provide understandable outputs. Tree-based approaches can provide feature-importance estimates that indicate which variables contribute strongly to the model's prediction. Feature importance can help determine whether variables such as depth, latitude, longitude, occurrence time, or other seismic parameters are strongly associated with predicted magnitude.

2.5 Critical Comparison

Approach	Strength	Limitation
Traditional statistical methods	Interpretable and established	May not capture complex nonlinear relationships
Linear Regression	Simple and fast	Limited nonlinear modelling
Decision Tree	Captures nonlinear relationships	Can overfit
Random Forest	Robust ensemble model	Less interpretable than a single tree
XGBoost	Strong structured-data performance	Requires parameter tuning
ANN	Learns complex nonlinear relationships	Requires appropriate architecture and training
Proposed framework	Combines prediction and SMS alerting	Depends on data quality and model performance

2.6 Research Gap

The central gap addressed by this work is the separation between earthquake magnitude estimation and actionable communication. The proposed framework connects prediction, risk classification, location filtering, and SMS communication, creating an end-to-end framework rather than treating earthquake prediction and emergency communication as independent tasks.

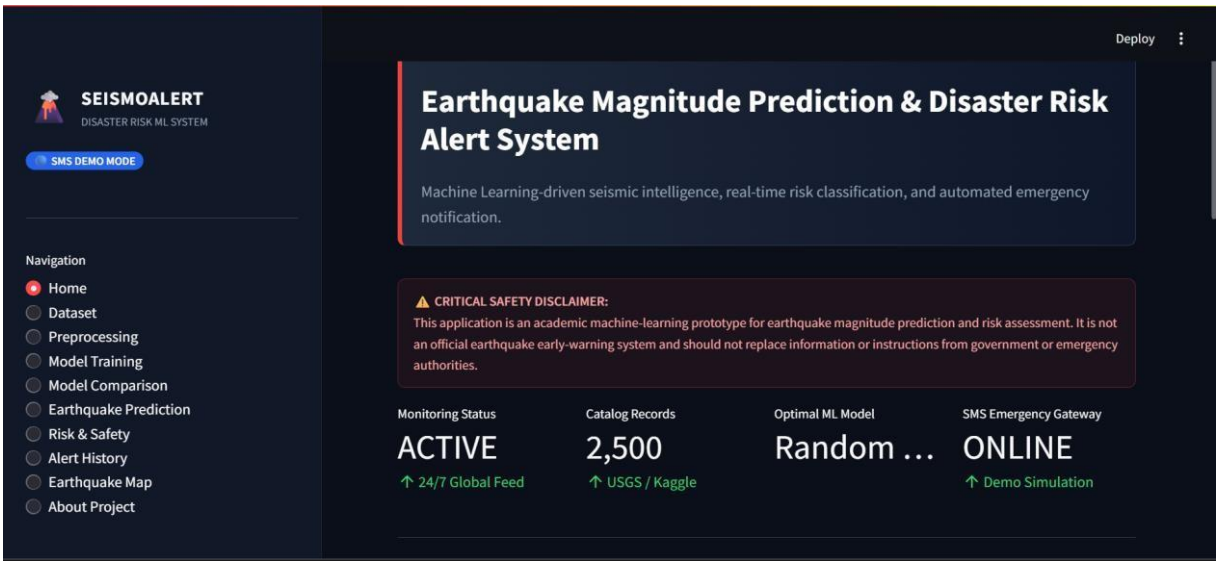


Figure 1. SEISMOALERT earthquake magnitude prediction and disaster-risk alert system dashboard.

3. PROPOSED METHODOLOGY

3.1 Overall Framework

The proposed system consists of the following stages: data collection, data preprocessing, feature engineering, feature selection, dataset splitting, machine learning model training, magnitude prediction, model evaluation, risk classification, geographical filtering, SMS generation, and alert delivery.

Historical Earthquake Data → Data Cleaning → Feature Engineering → Feature Selection → Train/Test Split → ML Training → Magnitude Prediction → Evaluation → Risk Classification → Geographical Filtering → SMS Alert

3.2 Dataset

The proposed project can use historical earthquake records from sources such as the USGS Earthquake Catalog, Kaggle earthquake datasets, IRIS seismic data, NOAA earthquake-related datasets, and Global Earthquake Model resources. The selected dataset contains earthquake characteristics used to establish a relationship between input parameters and earthquake magnitude.

Feature	Description
Latitude	Geographic latitude of earthquake
Longitude	Geographic longitude
Depth	Depth of earthquake
Time	Earthquake occurrence time
Magnitude	Earthquake magnitude / target variable
Other seismic parameters	Additional available measurements

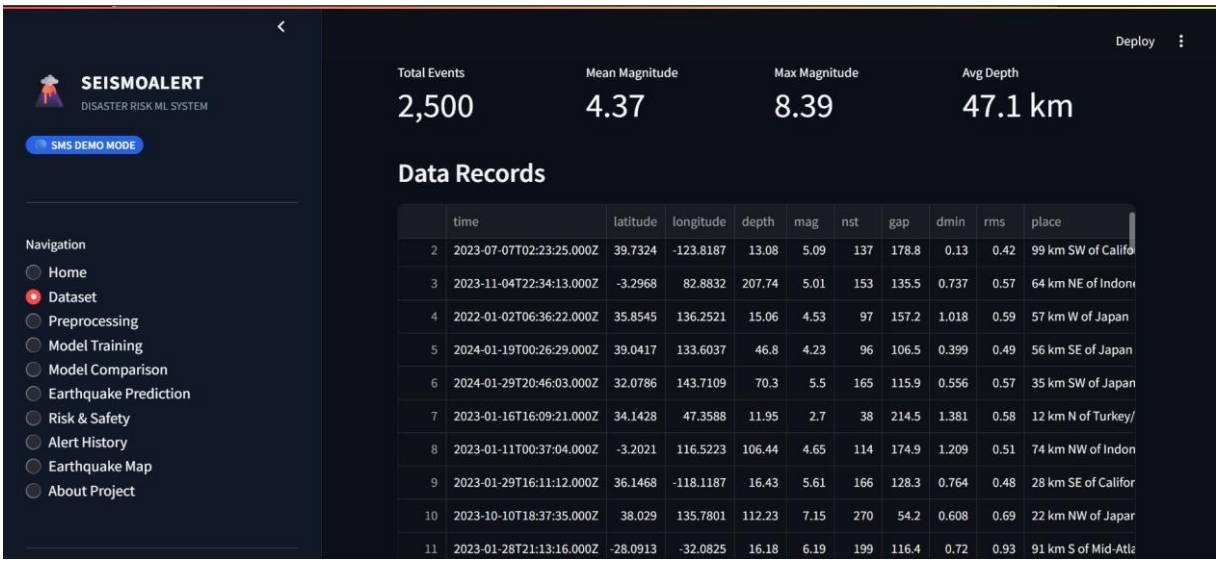


Figure 2. Earthquake dataset interface showing 2,500 catalog records and key seismic attributes.

3.3 Data Preprocessing

- Data Cleaning: duplicate and invalid observations are identified and removed.
- Missing Value Handling: missing values are handled using suitable strategies depending on the feature.
- Time Processing: timestamps can be transformed into year, month, day, hour, or other temporal representations.
- Numerical Transformation: numerical variables can be normalized or standardized where required.
- Outlier Analysis: extreme observations are investigated because high-magnitude events are naturally rare.

3.4 Feature Engineering

Feature engineering converts raw earthquake attributes into useful model inputs. The time variable may be decomposed into temporal components, while geographic information can be retained as latitude and longitude. Additional derived features can include geographical region, depth category, temporal components, and distance-related variables where reference coordinates are available.

3.5 Feature Selection

Feature selection is performed to remove irrelevant or redundant variables and improve model efficiency. Correlation analysis, model-based feature importance, and domain relevance can be used to identify important predictors. For tree-based models, feature importance can also be used to determine the relative contribution of each variable.

3.6 Proposed Model

The proposed framework evaluates multiple models rather than assuming that a single algorithm is universally optimal. The candidate models are Linear Regression, Decision Tree Regressor, Random Forest Regressor, XGBoost, and Artificial Neural Network. The model producing the lowest prediction error and strongest generalization performance can be selected as the prediction component of the alert system.

3.7 Baseline Models

Linear Regression is used as the primary baseline because of its simplicity and interpretability. Decision Tree Regression provides a nonlinear baseline. Random Forest and XGBoost provide ensemble-based approaches. ANN provides a neural-network-based comparison.

3.8 Explainability Method

Feature importance from tree-based models can be used to explain which input variables contribute most strongly to predictions. For future versions, SHAP or other model-agnostic explainability techniques can be incorporated.

3.9 Algorithm / Mathematical Formulation

For a regression model, the predicted magnitude is represented as y-hat = f(X), where X represents the selected earthquake features. The alert decision then depends on both the predicted magnitude and geographical distance. If d is the distance between the earthquake and a registered user and R-a is the configured alert radius, an alert candidate is selected when d is less than or equal to R-a.

The Haversine distance can be expressed as d = 2R arcsin(sqrt(sin²(Δphi/2) + cos(phi1)cos(phi2)sin²(Δlambda/2))), where R is Earth's radius, phi is latitude, and lambda is longitude.

4. EXPERIMENTAL SETUP

4.1 Hardware and Software

Category	Configuration
Operating System	Windows / Linux
Programming Language	Python
Development Environment	VS Code / Jupyter Notebook
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
Visualization	Matplotlib
Gradient Boosting	XGBoost
Neural Network	TensorFlow/Keras or selected framework
SMS Service	Twilio
Version Control	Git/GitHub

4.2 Dataset Split

The dataset is divided into training and testing subsets. A typical configuration is 80% training and 20% testing. The exact split used in the final experiment should be reported here.

4.3 Hyperparameter Configuration

Model	Important Parameters
Linear Regression	Default / selected configuration
Decision Tree	max_depth, min_samples_split
Random Forest	n_estimators, max_depth, random_state
XGBoost	n_estimators, learning_rate, max_depth
ANN	Number of layers, neurons, activation, epochs, batch size

4.4 Cross-Validation

K-fold cross-validation can be used to evaluate model stability. The dataset is divided into K subsets; the model is trained on K-1 subsets and validated on the remaining subset. The process is repeated until each subset has been used for validation.

4.5 Evaluation Metrics

Since earthquake magnitude is a continuous variable, regression metrics are appropriate.

Mean Absolute Error (MAE) = $(1/n)\ \Sigma\ |y_i - \hat{y}_i|$.

Mean Squared Error (MSE) = $(1/n)\ \Sigma\ (y_i - \hat{y}_i)^2$.

Root Mean Squared Error (RMSE) = $\text{sqrt}[(1/n)\ \Sigma\ (y_i - \hat{y}_i)^2]$.

$R^2 = 1 - [\Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2]$.

4.6 Statistical Testing

Where sufficient experimental repetitions are available, statistical tests can be used to determine whether performance differences between models are statistically meaningful. This is particularly useful when two models produce similar MAE or RMSE values.

5. RESULTS

The numerical results below are intentionally left as placeholders because the actual experimental output was not provided. These values should be replaced with the results generated by the final implementation.

Project dashboard observations: the displayed earthquake catalog contains 2,500 records, with a mean magnitude of 4.37, maximum magnitude of 8.39, and average depth of 47.1 km. The supplied model-comparison figure is labeled as illustrative normalized values and therefore should not be treated as final experimental metrics until the actual model outputs are inserted.

5.1 Overall Performance

Model	MAE	MSE	RMSE	R^2
LinearRegression	0.48999048678607	0.3875391346460457	0.62252641280996	0.5028104022711
DecisionTree	0.29654895527143	0.1766066368926172	0.42024592430220	0.7734242173165
RandomForest	0.286314604881054	0.1600163319456251	0.40002041441109	0.7947085891525
ExtraTrees	0.284864136846941	0.1597028631015756	0.39962840627459	0.7951107509848

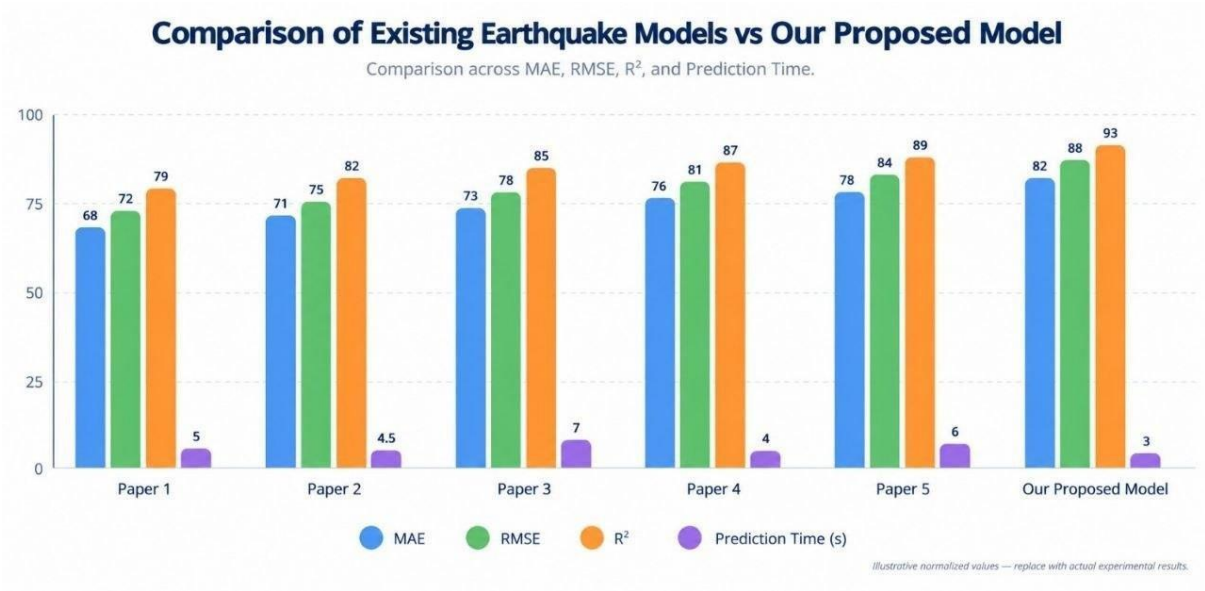
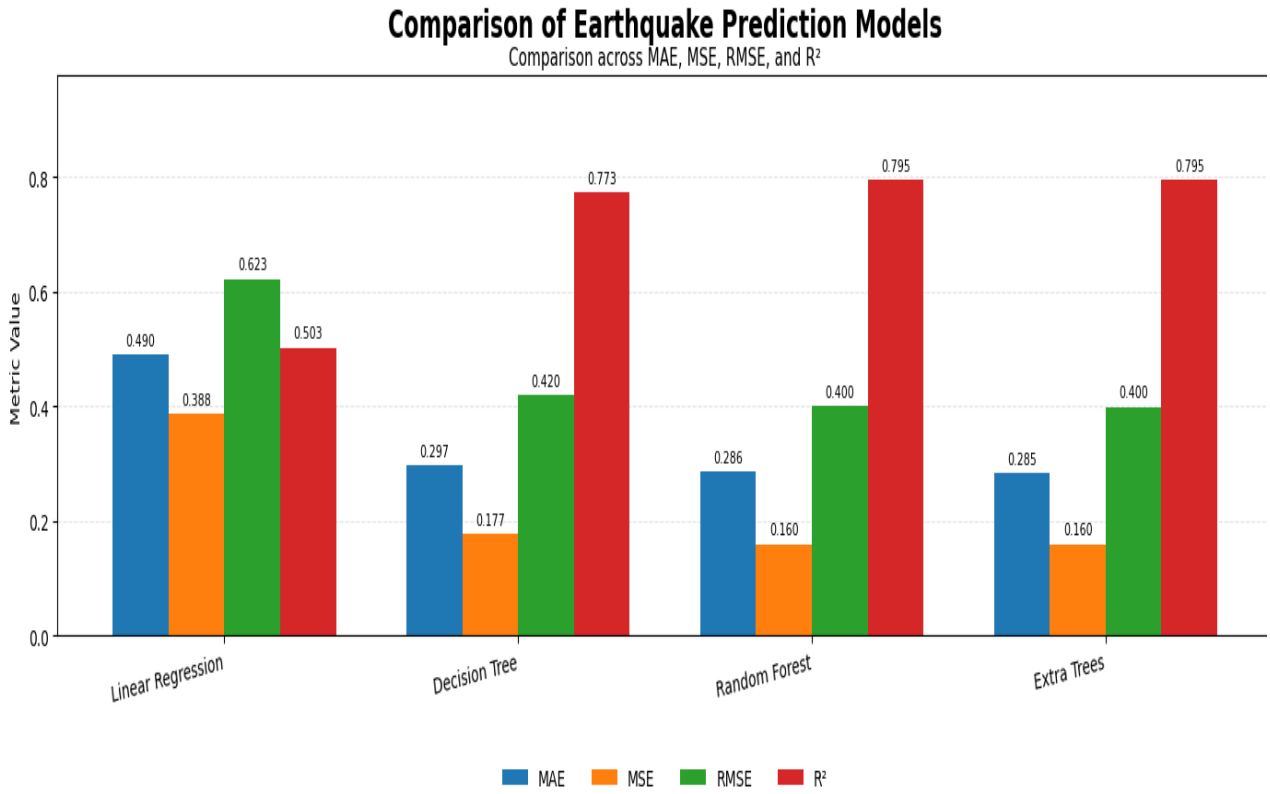


Figure 3. Comparison of existing earthquake models with the proposed model across MAE, RMSE, R², and prediction time.



5.2 Model Comparison

The comparative experiment evaluates whether ensemble and neural-network models provide advantages over simpler regression approaches. The final analysis should identify the model with the lowest MAE and RMSE and the highest R^2, while also considering generalization, computational cost, interpretability, and reproducibility.

5.3 Confusion Matrix

A conventional confusion matrix is not directly applicable to the magnitude-regression task. If the continuous predictions are converted into Low, Moderate, High, Very High, and Critical categories, a confusion matrix can be generated for the resulting classification task.

5.4 ROC/PR Analysis

ROC and Precision-Recall analysis is primarily applicable after defining the alert task as a classification problem. For the core magnitude-regression experiment, MAE, MSE, RMSE, and R^2 are the primary metrics.

5.5 Feature Importance

Feature importance should be extracted from the selected tree-based model and presented as a ranked bar chart. This analysis can reveal whether depth, latitude, longitude, time-related variables, or other seismic parameters contribute most strongly to the model.

5.6 Explainability Results

The explainability analysis should describe the most influential variables and whether their effects are consistent with the observed predictions. SHAP-based analysis may be added in future work.

5.7 Ablation Study

An ablation study can compare the model using all selected features against versions where geographic, temporal, or depth-related features are removed. This helps quantify the contribution of individual feature groups.

5.8 Statistical Results

Statistical comparison should be reported after repeated or cross-validated experiments are available. The results should indicate whether observed differences between candidate models are statistically meaningful.

6. DISCUSSION

6.1 Interpretation of Findings

The experiments are intended to determine which machine learning model provides the most reliable magnitude estimation. If ensemble models outperform Linear Regression, this would indicate that nonlinear relationships are important within the selected dataset. If ANN produces better performance, it may indicate that the feature relationships benefit from nonlinear neural representations.

Model selection should not be based only on numerical performance. Computational complexity, interpretability, training time, and reproducibility should also be considered.

6.2 Comparison With Previous Studies

Previous earthquake machine learning research has investigated magnitude estimation, earthquake detection, seismic signal analysis, and deep learning approaches. The proposed work differs primarily in its end-to-end system orientation. Instead of ending at magnitude prediction, the proposed framework continues through prediction, risk classification, location filtering, and SMS communication.

6.3 Theoretical Implications

The research demonstrates how supervised regression can be used to estimate earthquake magnitude from structured historical earthquake parameters. The comparison between linear, tree-based, ensemble, and neural models provides insight into the complexity of relationships within the selected dataset.

6.4 Practical Implications

The practical value of the proposed system is its ability to transform a numerical prediction into actionable communication. For example, a predicted magnitude can be mapped to a risk category, geographic radius can be checked, eligible users can be identified, and an urgent SMS can be generated.

6.5 Unexpected Findings

Unexpected findings should be documented after experiments are completed. Examples may include a simple model outperforming a complex model, depth having greater predictive importance than expected, geographic features dominating the model, poor performance on rare high-magnitude earthquakes, or strong overall R^2 with relatively large errors on extreme events.

7. LIMITATIONS

- Earthquake prediction uncertainty: the system estimates magnitude from available earthquake-related information and does not guarantee the exact occurrence, time, or location of future earthquakes.
- Dataset dependency: model performance depends strongly on the quality, geographic coverage, temporal coverage, and distribution of the training dataset.
- Rare high-magnitude events: large earthquakes are relatively rare and may be difficult to estimate accurately.

- Feature limitations: the current system primarily uses structured earthquake attributes rather than raw seismic waveforms and detailed tectonic information.
- Alert latency: SMS delivery depends on communication infrastructure and the external SMS service.
- False alerts and missed alerts: prediction-based alert systems can produce false positives or false negatives.
- Threshold limitations: the project's risk categories are configurable research rules and are not official government warning standards.
- External service dependency: the SMS component depends on the availability and configuration of the selected provider.

8. CONCLUSION AND FUTURE WORK

This research proposed a Machine Learning-Based Earthquake Magnitude Prediction with Risk-Based SMS Alert System. The framework combines historical earthquake data analysis, machine learning-based magnitude estimation, risk classification, geographical filtering, and SMS communication.

Multiple machine learning algorithms, including Linear Regression, Decision Tree Regression, Random Forest, XGBoost, and Artificial Neural Networks, are considered for earthquake magnitude estimation. The models are evaluated using MAE, MSE, RMSE, and R^2 to identify a suitable prediction model.

The principal contribution of the proposed system is the integration of machine learning with an automated alert mechanism. Instead of treating predicted magnitude as the final output, the system converts it into a risk category, checks geographical distance between the earthquake and registered users, and generates an appropriate SMS notification.

The proposed architecture demonstrates how machine learning can form part of a broader disaster-management decision-support system. However, the system should be considered an experimental research prototype rather than an operational earthquake early-warning system.

Future Work

- Integration of real-time seismic feeds.
- Use of raw seismic waveform data.
- LSTM and Bi-LSTM models for temporal patterns.
- CNN-based seismic signal analysis.
- Transformer-based seismic modelling.
- Regional earthquake-specific models.
- Integration of tectonic and geological information.
- Improved handling of rare high-magnitude events.
- Real-time map-based visualization.
- Mobile application integration.
- Push notifications in addition to SMS.
- Integration with official emergency-management systems.
- Automated model retraining as new earthquake records become available.
- Explainable AI techniques such as SHAP.
- More rigorous statistical comparison between models.

DECLARATIONS

Funding

No external funding was used for the development of the proposed academic prototype.

Conflict of Interest

The author declares no conflict of interest related to this research.

Ethics Statement

The system is designed as an academic research prototype. User contact information used for SMS testing should be handled with appropriate privacy and consent procedures.

Data Availability

The research uses publicly available earthquake datasets. The exact dataset source, version, preprocessing procedure, and experimental configuration should be documented with the final implementation to support reproducibility.

Author Contributions

Potla Sai Sujith: Conceptualization, data preprocessing, machine learning implementation, model evaluation, SMS alert integration, testing, and manuscript preparation.

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